# Shahan Ayyubi

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## Education

**Jamia Millia Islamia**, New Delhi 2024 – 2028
*Bachelor of Technology in Computer Engineering (CGPA: **8.81**/10)*

## Skills

**Languages:** Python, C++, Java, TypeScript, JavaScript, SQL, Bash
**ML & Inference:** PyTorch, Transformers, vLLM, ONNX, BladeDISC, Scikit-learn, NumPy, Pandas
**Vision & Robotics:** OpenCV, Visual Odometry, EKF, MPPI, Semantic Segmentation
**Systems & Data:** Docker, AWS, GCP, Linux, GitHub Actions, DuckDB, LanceDB, MongoDB
**Web:** React, Node.js, Express, Django, Three.js, Tailwind CSS

## Experience

**Research Intern**, *Jamia Millia Islamia, New Delhi* Sep 2026 – Present
- Applying computer vision to multi-pass friction stir processing of ZE41 Mg alloy, mapping deformation history across 7 overlapping passes (88 zones).
- Building a zone-recognition model fusing pass geometry with micrograph texture ($\mu$SAM, MatSAM); automating grain and particle analysis on $\sim$240 micrographs.

**Research Intern**, *Indraprastha Institute Of Information Technology, Delhi* Jun 2026 – Jul 2026
- Benchmarked HuggingFace vs vLLM (PagedAttention) on Qwen2-1.5B on a 4 GB GPU, reaching 2.9× throughput with INT4-AWQ and 27% less VRAM.
- Reproduced DVM and BladeDISC dynamic-shape ML compilers on BERT-base, showing 70–220× recompile overhead and 1.09–1.56× speedups across 8 shapes.

**Organiser** (prev. Cloud Lead), *Google Developer Groups on Campus, JMI* Aug 2025 – Present
- Promoted to Organiser in Aug 2026; managing all chapter teams and their leads.
- As Cloud Lead, ran 2 GC workshops, and a record Study Jam (150+, 2× YoY).

**Founder**, *teamV (Creative Agency)* Jul 2024 – Jan 2026
- Managed a 5-person freelance team, delivering 30+ marketing projects for 10+ clients, generating INR 3L+ in revenue while owning the end-to-end project lifecycle.

## Projects

**METAGROSS** | *Python, OpenCV, PyTorch, Three.js, AWS SageMaker* | Repository Aug – Sep 2026
- Built a vision-only autonomy stack for a GPS-denied ground robot (SIH 2026, BEL): stereo VO, EKF fusion, MPPI planning and a closed-loop simulator.
- Achieved 1.87% drift over 5.81 km of real KITTI drives at 31 Hz on CPU; trained a VO-failure detector (AUROC 0.957) and a terrain segmenter (0.83 mIoU).

**Cassetto** | *Python, tree-sitter, DuckDB, LanceDB* | Repository | PyPI May 2026
- Built a local code intelligence engine parsing 13 languages via tree-sitter into a call graph, vector store and keyword index, exposing 18 tools over MCP.
- Implemented hybrid retrieval (BM25 + semantic vectors) with Reciprocal Rank Fusion and graph-aware reranking, achieving 92% precision and 91% recall.
- Engineered graph analysis tools with sub-30ms latency; published to PyPI with cross-platform CI/CD across 3 OSes and 3 Python versions.

**Project Canary** | *PyTorch, CNN, BiLSTM, Attention* | Repository Aug – Oct 2025
- Built a CNN-BiLSTM with temporal attention fusing Sentinel-5P satellite imagery, ERA5 weather and CPCB station data to forecast ground-level $NO_2$ and $O_3$.
- Achieved $R^2$ of 0.76 ($NO_2$) and 0.67 ($O_3$) on held-out data, beating an XGBoost baseline by $\sim$10% in RMSE.

## Achievements

**Amazon ML Challenge 2026:** Ranked in the top 4% of teams
**LeetCode:** 600+ problems solved (top 15% globally)
**Certifications:** CS50x (Harvard), Google Cloud Study Jam, DevOps Essentials (IBM)